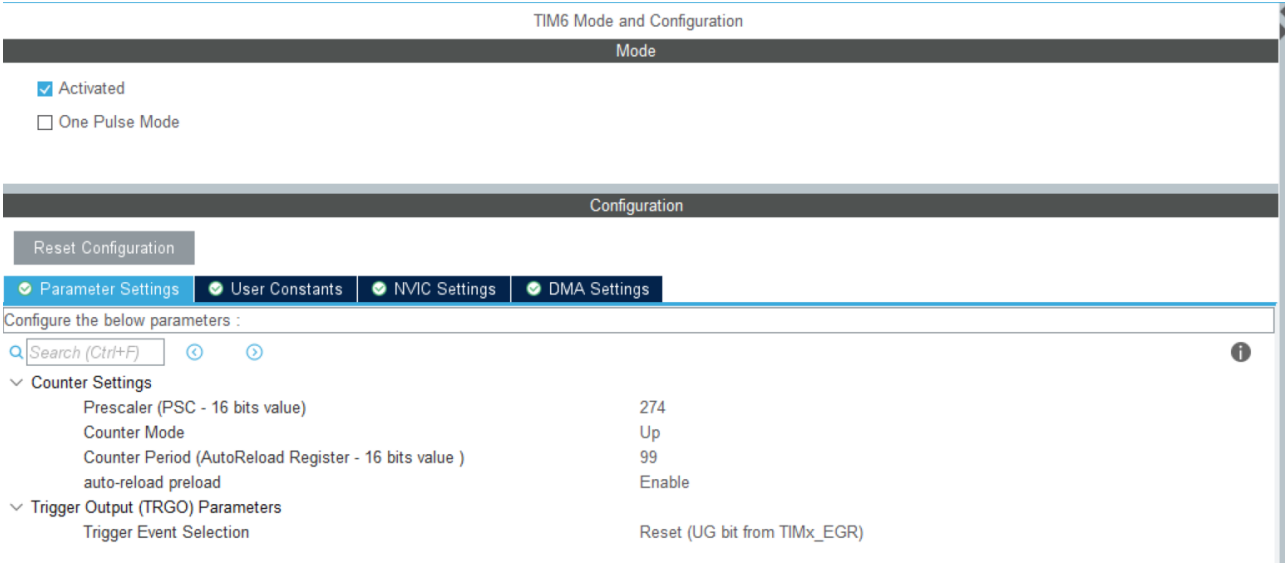


1. 文档说明

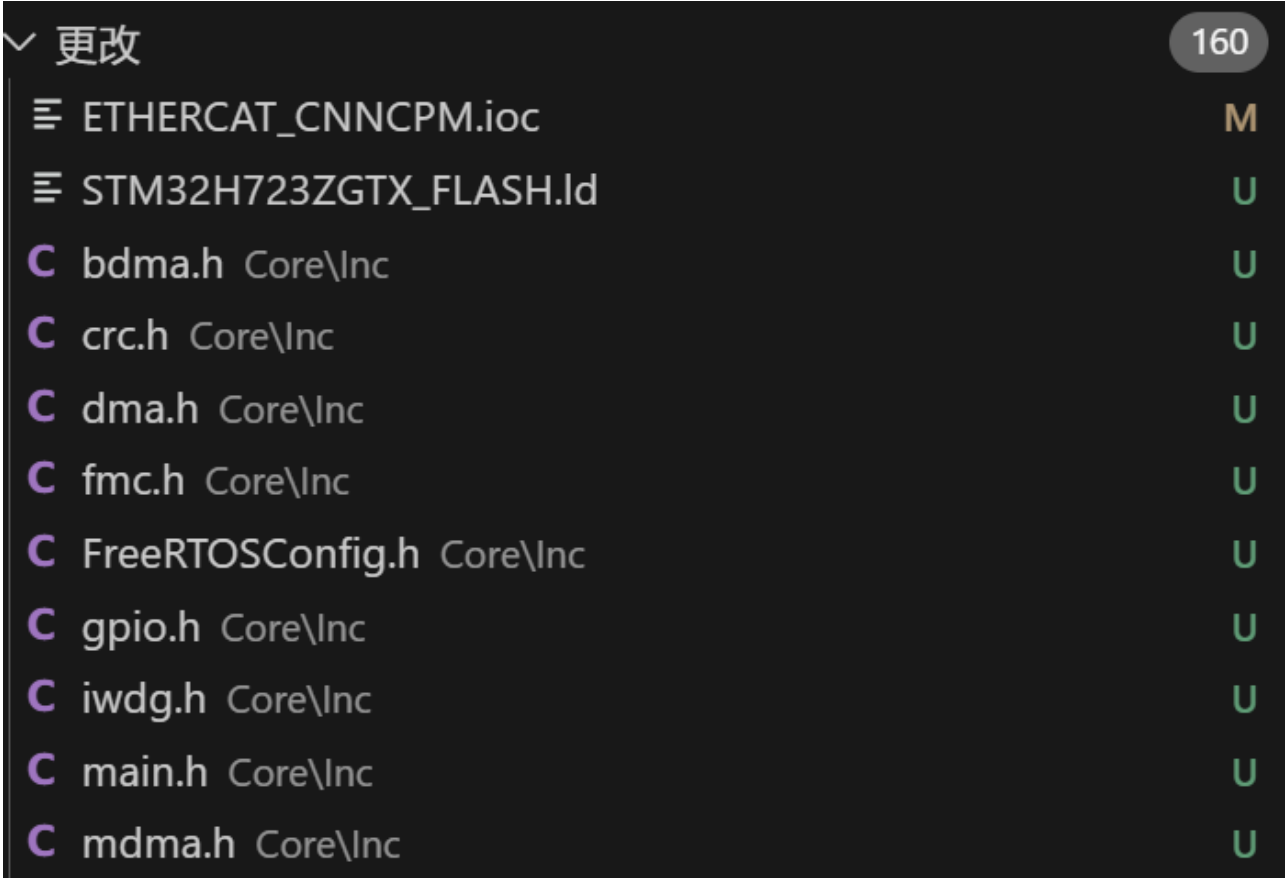
- 本demo针对新增片上外设，且不需要设备驱动情况时，工程修改参考
- 本demo拟实现如下功能：
 - 使用TIM6，中断周期为0.1ms，从而提供时间基准供FreeRTOS进行任务运行耗时统计
 - TIM6不占用GPIO，且不被业务需求依赖，故按工程拟定的原则，需要将其初始化放置到main函数内部
 - 基础分支hash: 8f895773ff55a79bed5662de1b436ddf827d4cd4

2. 修改方式

- 使用STM32CubeMX构建TIM6配置



- 生成工程，并筛选TIM6相关代码文件



C	octospi.h	Core\Inc	U
C	rtc.h	Core\Inc	U
C	spi.h	Core\Inc	U
C	stm32h7xx_hal_conf.h	Core\Inc	U
C	stm32h7xx_it.h	Core\Inc	U
C	tim.h	Core\Inc	1, U
C	usart.h	Core\Inc	U
C	bdma.c	Core\Src	U
C	crc.c	Core\Src	U
C	dma.c	Core\Src	U
C	fmc.c	Core\Src	U
C	freertos.c	Core\Src	U
C	gpio.c	Core\Src	U
C	iwdg.c	Core\Src	U
C	main.c	Core\Src	U
C	mdma.c	Core\Src	U
C	octospi.c	Core\Src	U
C	rtc.c	Core\Src	U
C	spi.c	Core\Src	U
C	stm32h7xx_hal_msp.c	Core\Src	U
C	stm32h7xx_it.c	Core\Src	U
C	system_stm32h7xx.c	Core\Src	U
C	tim.c	Core\Src	U
C	usart.c	Core\Src	U
🔑	LICENSE.txt	drivers\CMSIS	U

- 将步骤2中文件与原文件对比，差异部分拷贝到相应的原文件中

main.c

```
/* Initialize all configured peripherals */
MX_GPIO_Init();
MX_DMA_Init();
MX_MDMA_Init();
MX_BDMA_Init();
MX_FMC_Init();
MX_TIM1_Init();
MX_TIM2_Init();
MX_CRC_Init();
MX_IWDG1_Init();
MX_RTC_Init();
+ MX_TIM6_Init();
```

stm32h7xx_it.c

```
/* USER CODE END 0 */

/* External variables -----
extern TIM_HandleTypeDef htim1;
extern TIM_HandleTypeDef htim6;
/* USER CODE BEGIN EV */

/**
 * @brief This function handles TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts.
 */
void TIM6_DAC_IRQHandler(void)
{
    /* USER CODE BEGIN TIM6_DAC_IRQn 0 */

    /* USER CODE END TIM6_DAC_IRQn 0 */
    HAL_TIM_IRQHandler(&htim6);
    /* USER CODE BEGIN TIM6_DAC_IRQn 1 */

    /* USER CODE END TIM6_DAC_IRQn 1 */
}
```

tim.c

```
TIM_HandleTypeDef htim1;
TIM_HandleTypeDef htim2;
TIM_HandleTypeDef htim6;
```

```
/* TIM6 init function */
void MX_TIM6_Init(void)
{
    /* USER CODE BEGIN TIM6_Init 0 */

    /* USER CODE END TIM6_Init 0 */

    TIM_MasterConfigTypeDef sMasterConfig = {0};

    /* USER CODE BEGIN TIM6_Init 1 */

    /* USER CODE END TIM6_Init 1 */
    htim6.Instance = TIM6;
    htim6.Init.Prescaler = 274;
    htim6.Init.CounterMode = TIM_COUNTERMODE_UP;
    htim6.Init.Period = 99;
    htim6.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_ENABLE;
    if (HAL_TIM_Base_Init(&htim6) != HAL_OK)
    {
        Error_Handler();
    }
    sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
    sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
    if (HAL_TIMEx_MasterConfigSynchronization(&htim6, &sMasterConfig) != HAL_OK)
    {
        Error_Handler();
    }
    /* USER CODE BEGIN TIM6_Init 2 */

    /* USER CODE END TIM6_Init 2 */

}
```

```
void HAL_TIM_Base_MspInit(TIM_HandleTypeDef* tim_baseHandle)
{
    if(tim_baseHandle->Instance==TIM1) ...
    else if(tim_baseHandle->Instance==TIM2) ...
    else if(tim_baseHandle->Instance==TIM6)
    {
        /* USER CODE BEGIN TIM6_MspInit 0 */

        /* USER CODE END TIM6_MspInit 0 */
        /* TIM6 clock enable */
        __HAL_RCC_TIM6_CLK_ENABLE();

        /* TIM6 interrupt Init */
        HAL_NVIC_SetPriority(TIM6_DAC_IRQn, 5, 0);
        HAL_NVIC_EnableIRQ(TIM6_DAC_IRQn);
        /* USER CODE BEGIN TIM6_MspInit 1 */

        /* USER CODE END TIM6_MspInit 1 */
    }
}
```

You, 27秒钟前 • Uncommitted changes

```
void HAL_TIM_Base_MspDeInit(TIM_HandleTypeDef* tim_baseHandle)
{
    if(tim_baseHandle->Instance==TIM1) ...
    else if(tim_baseHandle->Instance==TIM2) ...
    else if(tim_baseHandle->Instance==TIM6)
    {
        /* USER CODE BEGIN TIM6_MspDeInit 0 */

        /* USER CODE END TIM6_MspDeInit 0 */
        /* Peripheral clock disable */
        __HAL_RCC_TIM6_CLK_DISABLE();

        /* TIM6 interrupt Deinit */
        HAL_NVIC_DisableIRQ(TIM6_DAC_IRQn);
        /* USER CODE BEGIN TIM6_MspDeInit 1 */
        You, 55秒钟前 • Uncommitted changes
        /* USER CODE END TIM6_MspDeInit 1 */
    }
}
```

tim.h

```
extern TIM_HandleTypeDef htim1;

extern TIM_HandleTypeDef htim2;

extern TIM_HandleTypeDef htim6;

/* USER CODE BEGIN Private defines */

/* USER CODE END Private defines */

void MX_TIM1_Init(void);
void MX_TIM2_Init(void);
void MX_TIM6_Init(void);
```

- 重新加载运行cmakelist.txt，以更新makefile文件

- 编译文件，结果如下

```

问题 5 输出 调试控制台 终端 端口 MEMORY XRTOS GITLENS

[ 86%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_pwr_ex.c.obj
[ 87%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rcc.c.obj
[ 88%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rcc_ex.c.obj
[ 89%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rtc.c.obj
[ 90%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_rtc_ex.c.obj
[ 91%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_sdram.c.obj
[ 92%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi.c.obj
[ 93%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi_ex.c.obj
[ 94%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim.c.obj
[ 95%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim_ex.c.obj
[ 96%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart.c.obj
[ 97%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart_ex.c.obj
[ 98%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_ll_fsmc.c.obj
[100%] Linking C executable ETHERCAT_CNNCPM.elf
Memory region      Used Size  Region Size  %age Used
ITCMRAM:           0 GB       64 KB         0.00%
DTCMRAM:           0 GB      128 KB         0.00%
FLASH:            161440 B       1 MB        15.40%
RAM_D1:            243704 B      320 KB        74.37%
RAM_D2:             1 KB        32 KB         3.13%
RAM_D3:             266 B        16 KB         1.62%
RAM_BKP:             6 B         4 KB         0.15%
Building F:\WORK\arm_core_develop\FW-ArmCore\ArmCoreV1_QSPI\build/ETHERCAT_CNNCPM.hex
Building F:\WORK\arm_core_develop\FW-ArmCore\ArmCoreV1_QSPI\build/ETHERCAT_CNNCPM.bin
[100%] Built target ETHERCAT_CNNCPM.elf
PS F:\WORK\arm_core_develop\FW-ArmCore\ArmCoreV1_QSPI\build>

```

- 添加代码：启动TIM6，并于中断中添加打印测试

```

/* TIM6 init function */
void MX_TIM6_Init(void)
{
    /* USER CODE BEGIN TIM6_Init 0 */

    /* USER CODE END TIM6_Init 0 */

    TIM_MasterConfigTypeDef sMasterConfig = {0};

    /* USER CODE BEGIN TIM6_Init 1 */

    /* USER CODE END TIM6_Init 1 */
    htim6.Instance = TIM6;
    htim6.Init.Prescaler = 274;
    htim6.Init.CounterMode = TIM_COUNTERMODE_UP;
    htim6.Init.Period = 99;
    htim6.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_ENABLE;
    if (HAL_TIM_Base_Init(&htim6) != HAL_OK)
    {
        Error_Handler();
    }
    sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
    sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
    if (HAL_TIMEx_MasterConfigSynchronization(&htim6, &sMasterConfig) != HAL_OK)
    {
        Error_Handler();
    }
    /* USER CODE BEGIN TIM6_Init 2 */
    HAL_TIM_Base_Start_IT(&htim6);
    /* USER CODE END TIM6_Init 2 */
}

```

```

/**
 * @brief This function handles TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts.
 */
void TIM6_DAC_IRQHandler(void)
{
    /* USER CODE BEGIN TIM6_DAC_IRQn 0 */
    printf("tim6 running\r\n");
    /* USER CODE END TIM6_DAC_IRQn 0 */
    HAL_TIM_IRQHandler(&htim6);
    /* USER CODE BEGIN TIM6_DAC_IRQn 1 */

    /* USER CODE END TIM6_DAC_IRQn 1 */
}

```

- 结果如下

[illegible]

- 最后，添加FreeRTOS统计功能函数及实现函数即可

```
/**
 * @brief This function handles TIM6 global interrupt, DAC1_CH1 and DAC1_CH2 underrun error interrupts.
 */
void TIM6_DAC_IRQHandler(void)
{
    /* USER CODE BEGIN TIM6_DAC_IRQn 0 */

    /* USER CODE END TIM6_DAC_IRQn 0 */
    HAL_TIM_IRQHandler(&htim6);
    /* USER CODE BEGIN TIM6_DAC_IRQn 1 */
    #ifdef configGENERATE_RUN_TIME_STATS
        run_time_count_increase();
    #endif
    /* USER CODE END TIM6_DAC_IRQn 1 */
}
```